

Möbius Reliability on the Twenty-Subgroup Lattice of a Named Abelian Core

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Abstract

We compute the exact generating-closure reliability law for sixteen frozen named coordinates in $Q \cong \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$. A label is deleted independently with probability q , and the retained labels generate a subgroup of the actual twenty-node subgroup lattice. For each subgroup H , we count the labels whose cyclic closure lies in H , form the cumulative probability $P_{\leq H}(q) = q^{16-n_H}$, and apply the integer Möbius function of that lattice. An independent enumeration of all $2^{16} = 65536$ supports agrees coefficient-by-coefficient for every subgroup. The top row is $1 - q - q^4 + q^5 - q^7 - q^8 + 5q^9 - 3q^{10}$, which factors as $(1 - q)(1 - q^4 - q^7 - 2q^8 + 3q^9)$ and reproduces the preceding C73 reliability result. The calculation is finite and source-bound; it makes no arithmetic, local, Euler-factor, root-number, Burnside-ring, or Hilbert–Polya claim.

1 Named finite probability object

Let

$$Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2, \quad L = \{S_1, \dots, S_{16}\}.$$

The sixteen coordinates and the ordered list of all twenty subgroups are inherited byte-for-byte from the C76 closure-orbit evidence. If $A \subseteq L$ is the retained support, write

$$\Phi(A) = \langle x_i : S_i \in A \rangle \leq Q, \quad \Phi(\emptyset) = \{0\}.$$

Every label is retained with probability $1 - q$ and deleted with probability q . Thus the exact probability of a closure event is a polynomial in the single formal variable q ; no floating-point evaluation is used.

For each label let $C_i = \langle x_i \rangle$. Define

$$n_H = \#\{i : C_i \leq H\}.$$

The labels outside this set must be deleted for the generated subgroup to be contained in H , while labels in the set are unrestricted.

Theorem 1 (Cumulative law). *For every one of the twenty source-ordered subgroups,*

$$P_{\leq H}(q) := \mathbb{P}[\Phi(A) \leq H] = q^{16-n_H}.$$

The C73–C76 source binding is

```
sha256(C73 evidence) = e91c8e6dcf1de5362b1a052ada83eb758b2c2d75520c1e8bdbd37ab055c725e5,
sha256(C73 manifest) = a50b5707d36f8b94b463e6c5fc4b5b7f6d6df7eb5e87d70bfc82d2b1a653cd8d,
sha256(C75 evidence) = 8beee17a227153e066907549df70c14a087b7de4141c3092d7cebd4a91541d98,
sha256(C75 manifest) = 7ede3e35c3101d17c683d2da440037d5bd4e002266530b52b3d1cb36ed4c8fcb,
sha256(C76 evidence) = 42e7783b2652666b84ea7f82b65d2421d98064ee5d5011ab94033aa18c051a94,
sha256(C76 manifest) = 55725664005113ae993b54197ff4fbd97bde347ce49aa69ea0c228372ba289d5.
```

2 Möbius inversion

Let $K \leq H$ denote containment in the actual subgroup poset, and let $\mu(K, H)$ be its integer Möbius function. The cumulative events satisfy $P_{\leq H} = \sum_{K \leq H} P_{=K}$, so inversion gives the exact identity

$$P_{=H}(q) = \sum_{K \leq H} \mu(K, H) q^{16-n_K}. \quad (1)$$

The source-ordered subgroup orders and containment counts are

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$ H_i $	1	2	3	3	3	3	6	6	6	6	9	9	9	9	18	18	18	18	27	54
n_{H_i}	5	6	7	6	7	6	8	7	8	7	11	7	7	8	12	8	8	9	15	16

For a direct check, let $c_{H,k}$ be the number of supports of size k with closure exactly H . Exhaustive enumeration yields

$$P_{=H}^{\text{dir}}(q) = \sum_{k=0}^{16} c_{H,k} (1-q)^k q^{16-k}. \quad (2)$$

Proposition 1 (Exact inversion check). *For all twenty subgroups, the integer coefficient vectors obtained from (1) and (2) are identical. The direct support totals by index are*

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\sum_k c_{H_i,k}$	32	32	96	32	96	32	96	32	96	32	1760	64	64	192	1760	64	64	192	30400	30400

These totals sum to 65536, and $\sum_H P_{=H}(q) = 1$ identically.

3 The complete polynomial atlas

For compactness, each row below lists the nonzero terms as “exponent: coefficient” pairs in q . This is the canonical integer coefficient representation of (1), not a numerical approximation.

index i	nonzero coefficients of $P_{=H_i}(q)$
0	11 : 1
1	10 : 1, 11 : -1
2	9 : 1, 11 : -1
3	10 : 1, 11 : -1
4	9 : 1, 11 : -1
5	10 : 1, 11 : -1
6	8 : 1, 9 : -1, 10 : -1, 11 : 1
7	9 : 1, 10 : -2, 11 : 1
8	8 : 1, 9 : -1, 10 : -1, 11 : 1
9	9 : 1, 10 : -2, 11 : 1
10	5 : 1, 9 : -2, 10 : -2, 11 : 3
11	9 : 1, 10 : -1
12	9 : 1, 10 : -1
13	8 : 1, 10 : -1
14	4 : 1, 5 : -1, 8 : -2, 10 : 5, 11 : -3
15	8 : 1, 9 : -2, 10 : 1
16	8 : 1, 9 : -2, 10 : 1
17	7 : 1, 8 : -1, 9 : -1, 10 : 1
18	1 : 1, 5 : -1, 8 : -1, 9 : -2, 10 : 3
19	0 : 1, 1 : -1, 4 : -1, 5 : 1, 7 : -1, 8 : -1, 9 : 5, 10 : -3

The final row corresponds to $H_{19} = Q$:

$$\begin{aligned} P_{=Q}(q) &= 1 - q - q^4 + q^5 - q^7 - q^8 + 5q^9 - 3q^{10} \\ &= (1-q)(1-q^4-q^7-2q^8+3q^9). \end{aligned} \quad (3)$$

Equation (3) is exactly the homogeneous reliability polynomial reported by C73. The agreement is a cross-generation consistency check: C77 derives it from the full subgroup lattice, while C73 obtained it from the generation criterion.

4 Reproducibility and scope

The producer binds all C73, C75, and C76 bytes before reading source data. An independent checker reconstructs cyclic closures, containment, the Möbius recursion, and all 65536 direct supports. Separate symbolic cross-check, clean replay, and hostile semantic mutations are required release gates. The effective C76 label action and its six-element ambient-kernel boundary are retained; no non-faithful ambient lift is substituted.

This theorem note concerns only a finite named presentation and its deletion probability polynomials. It does not claim a full table of marks or Burnside ring, an arithmetic or local object, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator. The scope firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.